

4



A man in a grey suit and tie is looking at a silver car. He is holding a white folder or document. The background is a bright, modern interior.

Working maths

How can you use what you will learn in this chapter?

- Check that the pay you receive from your job is correct.
- Choose the better job after comparing pay and conditions.
- Make sure that your employer is paying you the correct amount of superannuation.
- Determine the tax you have to pay.
- Read and interpret street maps.

Problem

David has just been offered a job selling cars. He has to choose between two different pay options.

Option 1

Work a 35-hour week at \$12.60 per hour.

Option 2

Receive a retainer of \$240 per week plus 2% commission on the total value of the weekly sales he makes.

Which pay option should David accept?

4.01

Wages and salaries



When you start working for an employer, it is important to know whether you are going to receive a wage or a salary.

If you earn a **wage**, you will be paid for every hour you work. The more hours you work, the more pay you will earn.

In some jobs, people who are paid a wage can work **overtime** at a higher rate of pay.

If you earn an **annual salary**, you will receive a fixed pay for the year irrespective of the number of hours you work. People employed 'on salary' are never paid for any hours of overtime they work.

Usually, people with university or professional qualifications earn a salary.

Practical activity: Jobs and pay investigation

In this activity you will be looking for information about a variety of jobs. You will need either Internet access or a copy of a newspaper's employment section.

What you have to do

- 1** Use the newspaper employment section or log onto a job website; for example, www.seek.com.au or www.mycareer.com.au.
- 2** Find five different types of jobs that you or your friends would like to have when you leave school. For each job, record:
 - whether the job is part-time or full-time, and permanent or casual
 - any qualifications or experience required for the job
 - whether the pay is a wage or a salary
 - the size of the pay
 - the hours you are required to work
 - how you apply for the job.
- 3** Make a summary of jobs based on their pay type. Find ten jobs that pay a salary and ten jobs that pay wages.

4.02

Earning a wage

Wage earners are paid for every hour they work.

Worked examples 4.02

Ali is an apprentice mechanic. He earns \$8.46 per hour. How much will Ali earn for working a 34-hour week?



Think

Ali earns \$8.46 for every hour he works. I have to multiply \$8.46 by the number of hours he works.

Write

$$\begin{aligned}\text{Ali's pay} &= \$8.46 \times 34 \\ &= \$287.64\end{aligned}$$

Last week Ali earned \$329.94. How many hours did he work?

Think

Ali earns \$8.46 every hour. If I divide the amount he earned by his pay each hour, I'll work out how many hours he worked.

Write

$$\begin{aligned}\text{Hours Ali worked} &= \$329.94 \div \$8.46 \\ &= 39 \text{ hours}\end{aligned}$$

Julia earns \$7.20 per hour in her part-time job. She works 3 hours on each of four afternoons per week. How much does Julia earn per fortnight?

Think

A fortnight is 2 weeks. I need to calculate the number of hours she works in 2 weeks, then multiply by the amount of money she earns each hour.

Write

$$\begin{aligned}\text{Hours per week} &= 3 \times 4 \\ &= 12 \text{ hours} \\ \text{Hours per fortnight} &= 12 \times 2 \\ &= 24 \text{ hours} \\ \text{Pay per fortnight} &= \$7.20 \times 24 \\ &= \$172.80\end{aligned}$$

Menhal's salary as a consultant in a fast-food company is \$85 000 per year. How much does Menhal earn per **fortnight**?

Think

There are 26 fortnights in a year. I have to divide her yearly pay by 26.

Write

$$\begin{aligned}\text{Fortnightly pay} &= \$85\,000 \div 26 \\ &= \$3269.23\end{aligned}$$

Exercise 4.02

It's true!

The minimum wage a 16-year-old can be paid is \$6.64 if they are a full- or part-time permanent employee, or \$7.96 if they are casual.

- 1 Yasmine works in a bakery. She earns \$12.24 per hour.
 - a How much does Yasmine earn for working an 8-hour day?
 - b Last week, Yasmine worked 36 hours. How much did she earn last week?
- 2 Sarah earns \$11.80 per hour delivering pizzas.
 - a How much does Sarah earn for working a 4-hour shift?
 - b Sarah thinks that there is a mistake in her pay advice. The advice states that she earned \$132.96 for working 12 hours. Has she been underpaid or overpaid? If she has, by how much?
- 3 Kyle works as an apprentice builder. His pay is \$16 per hour.
 - a How much does Kyle earn for working a $7\frac{1}{2}$ -hour day?
 - b Kyle works four $7\frac{1}{2}$ -hour days per week. How much does he earn per week?



HINT!

Remember, there are 52 weeks or 26 fortnights in a year. One fortnight is equivalent to 2 weeks.

- 4 Miranda earns \$562 per week. How much does she earn per fortnight?
- 5 Belinda is a computer programmer for a large newspaper company. Her annual salary is \$105 000. How much does Belinda earn per week?
- 6 Aaron's annual salary as a surveyor is \$96 000. How much does Aaron earn per fortnight?
- 7 John makes coffees and serves customers in a cafe. This table shows the hours he worked last week.

Day	Morning	Afternoon/evening
Sunday	8 am–12 noon	2 pm–5 pm
Tuesday	10 am–12 noon	1 pm–4 pm
Wednesday	9 am–12 noon	1 pm–3 pm
Thursday		4 pm–10 pm
Friday		3 pm–9 pm
Saturday	7 am–11 am	

- a How many hours did John work last Tuesday?
- b On which day did John work 7 hours?
- c How many hours did John work last week?
- d John's pay is \$14.34 per hour. How much did he earn last week?

- 8** This table shows the number of hours each employee at a coffee shop worked last week, and their rates of pay. Find the missing values.

Employee	Hourly rate of pay	Number of hours worked	Pay
Alex	\$18.50	25	a
Meshal	\$10.26	31	b
Toby	c	10	\$135
Carolyn	\$12	d	\$294
Robina	e	15.5	\$368.90

HINT!

School careers advisors can do more than provide you with a work experience job. They can help you to find part-time jobs and to determine the type of career to which you are suited.

- 9** Lachlan is a junior in a legal office. He is paid \$2296 per fortnight. How much does Lachlan earn per year?
- 10** Lewis works for a company that makes shade sails. He earns \$200 for working an 8-hour day. Lewis works 5 days per week for 42 weeks of the year. How much does Lewis earn per year?



- 11** Jewel works night shift in a nursing home, on 5 nights each week. Her shift begins at 2200 one night and finishes at 0700 the next morning.
- a** During her shift she has a 30-minute, unpaid meal break. How many hours does Jewel work per shift?
- b** Jewel earns \$1955 per fortnight. How much is Jewel paid per hour?

HINT!

Remember, in 24-hour time, am time is from 0000 (midnight) to 1200 (noon), and pm time is from 1200 to 2400 (midnight).

Practical activity: What's the right job for you?

To complete this activity you will need access to the Internet.

What you have to do

- Log into the site www.myfuture.edu.au and enter 'My guide'. The first time you use the website, you will need to register as a new user. Remember your name and password, as you will need to reproduce them exactly when you visit the site again.
- Complete some of the questionnaires in the 'Identifying' section. The questionnaires cover such things as your interests, abilities and whether you would like to work inside or outside.
- Ask the website to determine the types of jobs that are suited to your interests and abilities.
- Investigate some of the occupations in the careers market. Your perfect job may be something you have never considered.

4.03

Working overtime

When you are required to work more than your normal number of hours per day, or at a different time from your usual working hours, in most jobs you are paid extra per hour.

The common overtime rates are **time-and-a-half** and **double-time**.

Fact box

At time-and-a-half, the pay per hour is $1\frac{1}{2}$ times the normal rate.

At double-time, the pay per hour is double the normal pay rate.

Worked examples 4.03

Russell's normal pay rate is \$9.29 per hour. How much will he be paid for working 3 hours at double-time?

Think

When Russell works at double-time, he earns double the pay per hour. Each hour he'll earn $2 \times \$9.29 = \18.58 .



Write

Pay for working 3 hours at double-time
 $= 3 \times \$18.58$
 $= \$55.74$

Margot is employed to work from 8:30 am to 4 pm each day, with a 30-minute, unpaid lunch break. Her normal pay is \$11.20 per hour. For the first 3 hours of overtime she is paid time-and-a-half. Any extra hours she works are at double-time. Today Margot worked from 8:30 am until 9 pm. How much will she be paid?

Think

Normally, Margot is at work for $7\frac{1}{2}$ hours per day and she is paid for 7 hours. Today she was at work for $12\frac{1}{2}$ hours and she will be paid for 12 hours. Margot will receive normal pay for the first 7 hours. She will receive time-and-a-half for the next 3 hours. That makes 10 hours. For the 2 extra hours Margot will receive double-time.

Write

Normal pay = \$11.20 per hour
 Pay at time-and-a-half = $\$11.20 \times 1\frac{1}{2}$
 $= \$16.80$ per hour
 Pay at double-time = $\$11.20 \times 2$
 $= \$22.40$ per hour
 Today's pay
 $= 7 \times \$11.20 + 3 \times \$16.80 + 2 \times \$22.40$
 $= \$173.60$

Exercise 4.03

- 1 The table below shows some normal pay rates and pay rates at time-and-a-half and double-time. The first row has been completed as an example. Complete the missing parts in the table.

Normal pay rate	Pay rate at time-and-a-half	Pay rate at double-time
\$12	$\$12 \times 1\frac{1}{2} = \18	$\$12 \times 2 = \24
\$16	a	b
\$9.50	c	d
e	f	\$30

It's true!

Some of the highest-paid corporate bosses earn more than \$20 000 000 a year. That's equivalent to over \$9600 per hour.

- 2 Karin's normal rate of pay is \$9.60 per hour. When she works on public holidays, she is paid double-time. How much will Karin be paid for working 7 hours on a public holiday?
- 3 Grant is paid time-and-a-half for all work he does on Saturdays. His normal rate of pay is \$24.10 per hour. How much will he earn for working 4 hours on Saturday?
- 4 Neesha works in a service station. Her normal rate of pay is \$9.80 per hour. On Saturdays she receives time-and-a-half, and on Sundays she receives double-time. Calculate Neesha's pay for working 6 hours on Saturday and 3 hours on Sunday.
- 5 Guy is a trainer at the local gym. He earns \$17 per hour during the week, time-and-a-half on Saturdays and double-time on Sundays. The table shows the hours he worked last week. Calculate Guy's pay for the week.



WORKSHEETS

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Hours worked	6	9	5	10	6	4	3

- 6 Jarrod works as a tree lopper. He earns \$22 per hour for the first 38 hours that he works per week. When he works overtime, Jarrod is paid time-and-a-half for the first 5 hours overtime and double-time after that. Jason worked 52 hours in the week following a severe storm. Calculate his pay for that week.
- 7 For the same rate of pay, which work earns the higher amount of pay:
- 4 hours at time-and-a-half, or
 - 3 hours at double-time?
- Justify your answer.



4.04

Workplace allowances

Some employers pay **allowances** to employees who work in unpleasant or dangerous conditions or who are required to carry out special, unpleasant or dangerous duties.

Worked examples 4.04

Ken works for a pest control company. He is paid \$14.23 per hour. When he is required to handle chemicals containing cyanide, he is paid an additional \$2.82 per day. Last week, Ken worked 38 hours over 5 days and each day he handled chemicals containing cyanide. Calculate Ken's pay for last week.



Think

Ken will be paid for 38 hours at \$14.23 per hour. In addition he will be paid \$2.82 for each of the 5 days.

Write

$$\begin{aligned}\text{Ken's pay} &= 38 \times \$14.23 + 5 \times \$2.82 \\ &= \$554.84\end{aligned}$$

Katya cooks fish and chips in a take-away cafe. She is paid \$9.10 per hour. For any hours she works between 10 pm and 12 midnight, she receives an additional \$1.40 per hour. Katya's normal shifts are from 6 pm to midnight. How much does she earn on a normal shift?



Think

When Katya works between 10 pm and midnight, she earns $\$9.10 + \$1.40 = \$10.50$ per hour. On a normal shift, Katya works for 4 hours at \$9.10 per hour and 2 hours at \$10.50 per hour.

Write

$$\begin{aligned}\text{Katya's pay} &= 4 \times \$9.10 + 2 \times \$10.50 \\ &= \$57.40\end{aligned}$$

Exercise 4.04

- 1 Dwayne earns \$13.25 per hour as a cleaner. When he is required to clean toilets, he is paid an additional 85 cents per hour. Dwayne works a 35-hour week, which includes 5 hours of cleaning toilets.
 - a How much does Dwayne earn per hour when he cleans toilets?
 - b How many hours per week does Dwayne work cleaning non-toilet areas?
 - c Calculate Dwayne's weekly pay.
- 2 Madeleine is required to wear a white lab coat when she works in a pharmacy. She is paid \$12.65 per hour, and she receives a \$2.10 **laundry allowance** per day for her white coat. Each week, Madeleine works 32 hours over 4 days.
 - a How much is Madeleine paid per week in laundry allowances?
 - b Calculate Madeleine's weekly pay.
- 3 Luke delivers heavy items for an electrical store. He is paid \$14.75 per hour on Monday to Friday and time-and-a-half on weekends. In addition he receives \$2.95 per day for handling heavy items. The table shows the number of hours Luke worked each day last week. Use the information to determine Luke's pay for last week.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Hours worked	6	5	7	9	0	4	3

- 4 Justine works in a real-estate agency. She is paid \$682 per week, plus 55c per kilometre for every kilometre she drives in her car on real-estate business. Calculate Justine's pay for a week when she drove her car 163 km on real-estate business.
- 5 Daniel is a plumber's apprentice. He is paid \$8.55 per hour. In addition he receives 94 cents per hour when he is required to work in mud. On Thursday, Daniel worked for 7 hours, including 3 hours digging drains in the mud. How much did Daniel earn on Thursday?
- 6 Maurice is a security guard. He is employed to work a 5-hour shift on six nights per week. His job requires him to have a guard dog with him at all times. Maurice is paid \$15.28 per hour, \$61.25 per week for his guard dog and \$16.70 per shift for travel.
 - a How many hours does Maurice work per week?
 - b How much does Maurice receive per week for travel?
 - c Calculate Maurice's weekly pay.

It's true!

Australia was the first country to implement a minimum wage for workers. The minimum wage is based on the needs of workers, not on the ability of employers to pay.



4.05

Commission and retainers

People who sell items are often paid **commission**. Some workers receive only commission, while others receive a small amount plus commission. The small amount is called a **retainer**.

Worked examples 4.05

Michelle sells designer children's clothes. She receives 8% commission on all sales she makes. How much will Michelle receive from a home party where she sold children's clothes valued at \$1240?



Think

I need to calculate 8% of \$1240.
I can express 8% as either 0.08 or $\frac{8}{100}$.

Write

$$\begin{aligned}\text{Commission} &= 8\% \text{ of } \$1240 \\ &= \frac{8}{100} \times \$1240 \\ &= \$99.20\end{aligned}$$

Carla sells used cars. She receives a weekly retainer of \$180 and 3% commission on the value of the cars she sells. This week, Carla sold two cars for \$3590 each and one car for \$5695. Calculate Carla's pay for this week.

Think

I need to calculate the total value of the cars Carla sold. Then I need to determine 3% of the amount. I'll have to add Carla's retainer and commission.

Write

$$\begin{aligned}\text{Carla's sales} &= 2 \times \$3590 + \$5695 \\ &= \$12\,875 \\ \text{Carla's commission} &= \frac{3}{100} \times \$12\,875 \\ &= \$386.25 \\ \text{Carla's pay} &= \text{retainer plus commission} \\ &= \$180 + \$386.25 \\ &= \$566.25\end{aligned}$$

Exercise 4.05

1 Calculate the following amounts.

a 10% of \$760

b 8% of \$120

c 7.5% of \$1360

d 2% of \$875

e 1.5% of \$340

f $1\frac{1}{2}\%$ of \$550

WORKSHEETS



- 2 Jonas earns 4% commission on all his sales. Calculate his commission for sales totalling \$2560.
- 3 Amanda is a telemarketer. She receives 10% commission on the sales she makes. Today her sales totalled \$1820. How much commission did Amanda make today?
- 4 Samuel works for an investment company. When he gets customers to invest in the company's products, he receives $1\frac{1}{2}\%$ commission on the value of the investment. How much commission will he receive for two investments of \$600 000 each?
- 5 Alan sells pre-paid funerals. He receives 5% commission on his total sales. How much commission will Alan receive for selling five funerals valued at \$4270 each and one funeral valued at \$6200?
- 6 Roberta sells advertising in the regional farming magazine. She receives $2\frac{1}{2}\%$ commission on the value of the advertising she sells. The table shows the costs of advertising in the magazine.

Type of advertisement	Cost
Whole page	\$380
Half page	\$210
Quarter page	\$66
Small	\$32
Back cover	\$460

- a Roberta sold 12 half-page ads, 25 quarter-page ads, one back cover ad and 75 small ads. Calculate the total value of the ads Roberta sold.
 - b How much commission will Roberta earn from the sales?
- 7 Tamara is a stock and station agent. When she sells large rural properties, she receives 2% commission on the first \$1 million of the selling price and 1% on the remaining value. How much commission will Tamara earn for selling a rural property for \$1.8 million?
 - 8 Alistair works in an art gallery. He receives a retainer of \$290 per week plus 2% commission on the value of the sales he makes. Calculate Alistair's pay in a week when he sold art valued at \$36 000.
 - 9 Kristin is the agent for a band. The band pays her a retainer of \$1000 per month plus 1% commission on the payment the band receives for performances. During the year, Kristin booked six large concerts that each paid the band \$150 000, and 32 club performances valued at \$24 000 each. Calculate Kristin's pay for the year.

HINT!

Write $1\frac{1}{2}$ as 1.5.
It's quicker and has fewer keys to press when you enter it into your calculator.



HINT!

\$1 million
= \$1 000 000

4.06 Superannuation

Superannuation is a form of saving to provide you with an income after you retire from the workforce.

Government **legislation** requires all Australian employers to contribute to superannuation on behalf of those of their employees who are aged between 18 and 70 years and who earn more than \$450 per month. The minimum rate of superannuation is 9% of the employee's salary.

Worked example 4.06

Caleb works a 40-hour week and he earns \$14.90 per hour. How much superannuation is Caleb's employer required to pay into a superannuation fund on Caleb's behalf?

Think

Caleb's employer must pay 9% of Caleb's wage into superannuation. I have to calculate the amount Caleb earns per week, then determine 9% of the amount.

Write

$$\begin{aligned}\text{Caleb's pay} &= \$14.90 \times 40 \\ &= \$596\end{aligned}$$

$$\begin{aligned}\text{Caleb's superannuation} &= 9\% \text{ of } \$596 \\ &= \frac{9}{100} \times \$596 \\ &= \$53.64\end{aligned}$$

Exercise 4.06

It's true!

If you are less than 18 years old and work 30 hours per week or less, your employer is not required to pay superannuation for you.

- 1 Calculate the weekly superannuation payment for each of the following employees, at the rate of 9% of their weekly wage.

Employee	Weekly wage	Superannuation
Sarah	\$600	a
Mitch	\$485	b
Keith	\$1268	c
Sally	\$965	d

- 2 Australian law requires employers to pay a minimum of 9% of an employee's wage in superannuation. Kirrily earns \$720 per week, and her employer pays \$140 per fortnight into her superannuation account. Is Kirrily's employer paying the correct amount of superannuation? Justify your answer.
- 3 Susanne's employer pays 9% superannuation for her. Susanne's wage increased from \$560 to \$640 per week. By what weekly amount did her superannuation increase?

- 4** Caitlyn earns \$16.10 per hour for a 38-hour week. Caitlyn's employer pays 9% of her wage into a superannuation account.
- How much does Caitlyn earn per week?
 - How much superannuation does her employer pay for Caitlyn?
 - Australian law requires employers to pay superannuation into employees' accounts at least every 13 weeks. Calculate the amount Caitlyn's employer pays into her account every 13 weeks.
- 5** Russell is a builder's labourer. He is paid \$11.20 per hour. He receives an additional \$15 per day when he is required to work in the sun when temperatures are over 35°C.
- Calculate Russell's pay for a 39-hour week when the temperature on 4 days was over 35°C.
 - How much superannuation will Russell's employer have to pay at the rate of 9%?
- 6** Carol works for a company that encourages its employees to contribute to superannuation. In addition to the 9% compulsory superannuation, the company pays an additional dollar for every \$2 that employees pay into superannuation. Carol earns \$960 per week and she pays 10% of her salary into superannuation.
- How much does Carol pay into superannuation each week?
 - Determine the amount the employer pays into Carol's superannuation account each week.
 - Calculate the total amount that is contributed to Carol's superannuation account each year.
- 7** Evan has \$246 000 in his superannuation account, and the account earns 8% p.a. interest. Evan earns \$1680 per fortnight from his job in the bank. Each fortnight, Evan pays \$200 into superannuation and his employer contributes 9% of his salary.
- How much interest will be earned on \$246 000 in 1 year?
 - By how much will Evan's superannuation account grow in the next year?
- 8** Barry is retired and he is using his superannuation savings to provide him with a monthly income. He had \$460 000 in his superannuation account, and the account earns 7.5% p.a. interest.
- Calculate the amount of interest Barry's account will earn in 1 year.
 - Each year, Barry withdraws 4% of the total value of the account, including interest, to spend. He spends the same amount each month. How much money from his superannuation account does Barry spend each month?
 - Will Barry be able to continue withdrawing from his superannuation account at the same rate for the rest of his life? Give a reason for your answer.



It's true!

Financial advisors recommend that people need more than \$650 000 in their superannuation account, or the equivalent in property and shares, to be able to afford to retire comfortably.

4.07

Paying tax

The Australian Government provides many services for the community. Most of the money required for services such as education, hospitals, Medicare, transport, defence and social security comes from taxation. Most Australian workers pay **income tax** under the **pay as you go (PAYG)** system. Under PAYG, employers **deduct** tax from their employees' **gross pay**. Employees receive their **take-home pay**, which is also called their **net pay**.

Fact box

Net pay (take-home pay) = gross pay – tax – any other deductions

Worked example 4.07

Paddy earns \$18 per hour teaching children to swim. She works 3 hours on each weekday afternoon and 5 hours on Saturdays. Each week her employer deducts \$72 for income tax and \$20 for superannuation from her pay. What is Paddy's net or take-home pay?



Think

I have to calculate the amount Paddy earns each week, then subtract the tax and her contribution to superannuation.

Write

$$\begin{aligned}\text{Hours Paddy works} &= 5 \times 3 + 5 \\ &= 20 \text{ hours}\end{aligned}$$

$$\begin{aligned}\text{Paddy's gross pay} &= \$18 \times 20 \\ &= \$360\end{aligned}$$

$$\begin{aligned}\text{Net pay} &= \text{gross pay} - \text{tax} - \text{other} \\ &= \$360 - \$72 - \$20 \\ &= \$268\end{aligned}$$

Exercise 4.07

- 1 Peter's gross pay per week is \$540. Each week his employer deducts \$105 for PAYG tax and \$15 for health insurance. How much is Peter's weekly net pay?
- 2 Joel pays \$136 tax per week. How much tax does he pay per year?

- 3 Complete the missing values in the table.

Employee	Gross pay	Tax deducted	Net pay
Imran	\$540	\$73	a
Amanda	\$220	\$16	b
Baily	c	\$79	\$281
Shelby	\$1625	d	\$1193

- 4 Rowan has a part-time job as a dance instructor. Each week his gross pay is \$268. His employer deducts \$24 for PAYG tax and \$96 for car loan repayments. What is Rowan's take-home pay?
- 5 Nadia is paid **quarterly**. Her gross pay per quarter is \$5700, and her employer deducts \$728 in PAYG tax each quarter.
- a** What is Nadia's quarterly net income?
 - b** What is Nadia's annual income?
 - c** How much tax does Nadia pay per year?
- 6 Beatrice's gross weekly pay is \$500, and her employer deducts \$66 per week for tax. William's net monthly pay after paying \$256 in tax is \$1744. Which one of Beatrice or William has the greater net annual pay, and by how much?

HINT!

An amount that is paid 'quarterly' is paid every quarter of a year, or every 3 months.

Practical activity: Online tax

Have you ever wondered how an employer knows exactly how much tax to take out of an employee's pay? Some pay offices calculate tax, but a lot of employers simply use an online tax calculator! In this practical activity you are going to pretend that you are the employer and you are going to calculate tax 'the easy' way.

To complete this activity you will need Internet access.

What you have to do

- 1 Log onto the site http://www.ato.gov.au/scripts/taxcalc/calc_standard_hire.asp.
- 2 Type a name for your employee in the space for the Payee.
- 3 Use the default settings for all the questions and enter a weekly gross pay in the Gross Earnings box.
- 4 Select 'Calculate' to determine the amount of tax to deduct from the employee's gross pay.
- 5 Select 'Start Again'. Imagine that the same employee has not provided you with a **tax file number (TFN)**. Select 'No' for the question about supplying a tax file number, then determine the amount of tax you are required to deduct from the employee's gross pay.
- 6 Repeat the calculations for a variety of different incomes. Describe the effect that *not* supplying a tax file number has on the amount employers are required to deduct in tax.

4.08

Will you get a tax refund?

It's true!

There are serious penalties for people convicted of cheating on income tax. The penalties can include large fines and gaol sentences.

At the end of each financial year, all businesses and people who earn money are required to complete a **tax return**, which summarises the income earned during the year and the tax paid. The Australian Taxation Office (ATO) then calculates the exact amount of tax required on the income. If too much tax has been paid, the ATO will **refund** the excess amount. If insufficient tax has been paid, a bill for the extra tax is issued. The ATO allows some work- and income-related expenses to be deducted from the **gross** (total) **income**. Income tax is calculated on the **taxable income**; that is, on the income after any **allowable deductions** have been subtracted from the gross income.

Fact box

Taxable income = total income – allowable deductions
Income tax is calculated on taxable income.

It's true!

Taxable income groups and rates of taxation are regularly changed to keep up with the changing income levels of the population.

Tax tables

Every year the Australian Government publishes a **tax table** for the next financial year, as part of the annual Budget. The following table shows the tax rates for 2007–08. In addition to the tax shown in the table, taxpayers are required to pay a **Medicare Levy** of 1.5% of their taxable income.

Taxable income	Tax on this income
\$1–\$6000	Nil
\$6001–\$30 000	15c for each \$1 over \$6000
\$30 001–\$75 000	\$3600 plus 30c for each \$1 over \$30 000
\$75 001–\$150 000	\$17 100 plus 40c for each \$1 over \$75 000
\$150 001 and over	\$47 100 plus 45c for each \$1 over \$150 000

Worked examples 4.08

Jim's gross annual salary is \$89 520, and he has a total of \$7540 in allowable deductions from his income. How much is Jim's Medicare Levy?

Think

The Medicare Levy is 1.5% of taxable income. I must first determine Jim's taxable income by subtracting his allowable deductions from his gross income. Then I'll determine the 1.5%.

Write

$$\begin{aligned}
 \text{Taxable income} &= \$89\,520 - \$7540 \\
 &= \$81\,980 \\
 \text{Medicare Levy} &= \frac{1.5}{100} \times \$81\,980 \\
 &= \$1229.70
 \end{aligned}$$

Anna's taxable income is \$79 823, and her employer has deducted \$21 690 in PAYG tax.

- a** How much income tax should Anna pay for the year?
- b** How much tax should she pay in total?
- c** Has Anna paid too little or too much tax?
- d** How much does Anna owe the ATO, or how much refund does the ATO owe her?

Think

- a** Anna's taxable income is in the range \$75 001–\$150 000. The rule for calculating tax in this range is \$17 100 plus 40c for each \$1 over \$75 000. 40c is \$0.40. I need to subtract \$75 000 from her taxable income to calculate the number of dollars over \$75 000.
- b** I need to calculate Anna's Medicare Levy and add it to her income tax, to find how much tax she should pay altogether.
- c** Anna has paid \$21 690 but she only needed to pay \$20 226.55.
- d** Anna is entitled to a refund from the ATO because she has paid more tax than she needed to.

Write

Income tax
 $= \$17\,100 + \$0.40 \times (79\,823 - 75\,000)$
 $= \$17\,100 + \0.40×4823
 $= \$19\,029.20$

Medicare Levy = 1.5% of \$79 823
 $= 0.015 \times \$79\,823$
 $= \$1197.35$

Total tax payable = \$19 029.20 + \$1197.35
 $= \$20\,226.55$

Anna has paid too much tax.

Refund = taxpaid – tax payable
 $= \$21\,690 - \$20\,226.55$
 $= \$1463.45$

Exercise 4.08

- 1 Calculate the Medicare Levy on each of these taxable incomes.
 - a** \$28 650
 - b** \$132 698
 - c** \$17 985
- 2 Jane's gross income is \$43 200 and she has allowable deductions totalling \$5100.
 - a** What is Jane's taxable income?
 - b** What Medicare Levy does Jane have to pay?
- 3 Use the tax table opposite to answer the following questions.
 - a** For what taxable incomes is no tax payable?
 - b** What is the rule for calculating the tax on an income between \$30 001 and \$75 000?
 - c** For what taxable incomes is the **marginal rate** of taxation 45c?
 - d** Mike used the calculation $\$3600 + \$0.30 \times (45\,250 - 30\,000)$ to determine the amount of tax he had to pay on his taxable income. What is Mike's taxable income?

It's true!

Al Capone was the leader of organised crime in America in the 1930s. He was suspected of organising murders, prostitution, bank robberies and gang violence. However, the only crime for which police were able to convict him was income tax evasion.

- 4 Copy the table and complete the entries. The first row has been completed as an example.

Taxable income	Tax calculation	Tax
\$32 000	$\$3600 + \$0.30 \times (32\,000 - 30\,000)$	\$4200
\$78 980	a	b
\$101 000	c	d
\$21 640	e	f
\$166 580	g	h

- 5 Jake's normal wage is \$39 per hour for a 40-hour week. Each year he has allowable deductions totalling \$2800.
- Calculate Jake's normal taxable income.
 - What is the income tax on Jake's normal annual income?
 - Because the business was very busy, Jake was required to complete 124 hours of overtime at time-and-a-half. Calculate Jake's annual taxable income including overtime.
 - How much additional income tax does Jake have to pay because of the overtime?
 - What fraction of the money Jake earned in overtime did he have to pay in income tax?
- 6 Gillian's annual salary is \$69 800. She has money invested in an interest-bearing deposit that pays her \$1500 in interest each year. Gillian also receives \$21 000 in rent each year from an investment property she owns. She has allowable deductions totalling \$5200 from her job and another \$18 000 from her investment property.
- Calculate Gillian's taxable income.
 - How much income tax is Gillian required to pay?
- 7 Michael's taxable income is \$74 000 per annum.
- Calculate the income tax payable on \$74 000.
 - Michael decided to pay \$26 000 into superannuation under a salary-sacrificing arrangement. This means that his superannuation contributions are an allowable deduction. What is Michael's taxable income with salary sacrificing?
 - How much income tax will Michael save by salary sacrificing \$26 000 to superannuation?
- 8 When you sell an investment property that you have owned for more than a year, half of the **capital gain** (increase in value) is added to your income, and you have to pay tax on it. Melissa's taxable income from her job is \$76 000. She has also sold her rental property for a capital gain of \$160 000, so she must add \$80 000 to her taxable income. How much of the \$80 000 will Melissa pay in income tax?

It's true!

If you own an investment property, you have to include the rent you receive in your income, but you can claim all the expenses of owning the property as allowable deductions. The council rates, depreciation, maintenance, insurance and all the interest on the loan can be claimed.

Practical activity: Do you need to complete a tax return?

To complete this activity you will need Internet access.

What you have to do

- 1 Log onto the ATO website, www.ato.gov.au.
- 2 Go to 'Tax returns' on the left-hand side and select 'Tax return essentials' in the 'Start here' section.
- 3 Go to the 'Frequently asked questions' section on the right-hand side, then enter the section 'Do I need to complete a tax return?'

Some people who have an income are not required to complete (lodge) a tax return, and others are. Your task is to determine which of the following people are required to complete a tax return.

Gaye: Gaye is 18 years old and she receives an Austudy payment. Her total taxable income from Austudy and her part-time job is \$11 500.

Jason: Jason inherited some money from his grandparents and he earns interest on the money. In addition, Jason has a part-time, after-school job. Jason's taxable income is \$6200.

Bond: Bond receives an Abstudy allowance and he has a part-time job. Bond's total taxable income is \$9500.

Bodie: Bodie is 17 years old. He attends TAFE and he doesn't have a job. Bodie's parents have money invested in his name. His taxable income is \$1500.

Kim: Kim is a Chinese national. She is in Australia on a student visa to learn English. She has a part-time job in a Thai restaurant, and the restaurant owner deducts PAYG tax for her. Her taxable income is \$4500.

Nikky: Nikky is an Australian citizen. She met Kim at school and they work together in the same part-time job in the Thai restaurant. The restaurant owner deducts PAYG tax for Nikky also. Her taxable income is \$4500.

Dalby: After school, Dalby works part-time in his grandfather's business. Dalby's taxable income is \$4500, and his grandfather doesn't deduct any tax.



4.09

Different types of employment

It's true!

The minimum wage for an 18-year-old is \$1.86 per hour more for casual employment than for permanent employment. The extra money is to make up for not receiving holiday pay or sick pay.

The Australian job market is very different today compared to 30 years ago. Today, a large proportion of people have casual or part-time positions, and many young women juggle work, family and home responsibilities.

Permanent employees tend to be paid less per hour than casual employees because permanent employees receive benefits that casual employees do not.

Class discussion activity

Discuss the working conditions associated with each type of employment listed in the table, then complete each section with a Y for 'yes', N for 'no' or S for 'sometimes'. The column for casual employment has been completed as an example.

	Permanent salary	Permanent wage	Casual	Part-time	Contract or temporary
Job security			N		
Holiday pay			N		
Sick pay			N		
Paid overtime			S		
Superannuation			Y		
Promotion possibilities			S		
Regular hours			S		
Safe working environment			S		
Holiday loading			N		

Practical activity: Completing a tax return

To complete this activity you will need a copy of a tax return or the electronic version available on the ATO website. You will need a name and matching tax file number to use the online form.

What you have to do

- 1 Complete the tax return for an 18-year-old student who has a part-time job. The student earns \$172 per week for 40 weeks of the year. In addition, they receive \$420 from the bank in interest. The student has paid \$1050 in PAYG tax.
- 2 Determine the size of any refund or extra tax resulting from the tax return.

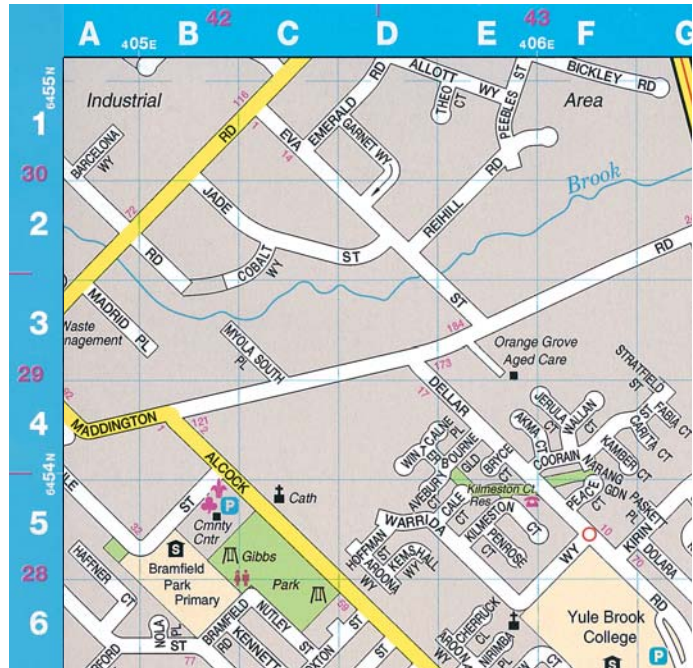
4.10

Trent's courier delivery job

Trent has just been employed as a courier delivery driver. As part of his job, he needs to make deliveries in the Perth suburbs. Trent needs to learn to read a street directory to help him locate his deliveries.

Locating positions on a street map

In Chapter 2 you learned about locating points on a graph. On a graph, the point (3, 1) means 3 on the horizontal axis and 1 on the vertical or side axis. Street maps use a similar system. Positions on a street map are given with a letter and a number; for example, A3. The letter refers to the horizontal axis and the number refers to the vertical axis.



Worked examples 4.10

What is the name of the aged care facility that is located at E3 on the map above?

Think

I have to find E on the horizontal 'axis' at the top of the page and look down the column. At the same time I have to find 3 on the side 'axis' and look across the row. The grid square where the row and column cross is position E3.

Write

On the map the aged care facility at E3 is called Orange Grove Aged Care.

What are the position coordinates of Gibbs Park?

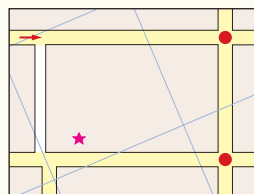
Think

Gibbs Park is located in columns B and C on the horizontal axis, and in rows 5 and 6 on the vertical axis.

Write

The location of Gibbs Park can be recorded as B5, B6, C5 or C6.

Trent noticed a pink star on a map. What does the pink star mean?



Think

Every book of road maps contains a page that explains what the symbols mean. A copy is on the facing page. I can find a pink star in the left-hand column.

Write

The pink star means a police station.

Exercise 4.10

- 1** Use the explanation of map symbols on the facing page to find the meanings of the following symbols.



Use the map on page 124 to answer questions **2** to **8**.

- 2** What features are at the following locations?

a J7

b K8

c F15

d J11

e J8

f C6

g F1

h L10

- 3** What are the grid references of the following places?

a The roundabout at the intersection of Albany Hwy and Austin Ave

b Kenwick railway station

c Hume Road Wildlife Reserve

d Thornlie Senior High School

- 4** How many sets of traffic lights are shown on Spencer Rd?

- 5** At N7 there is a black cross. What does the black cross mean?

- 6** How far is Maddington station from Perth?

- 7** Trent has to deliver a parcel to Liddelow Homestead at F3. In what avenue is the homestead?

- 8** Thornlie Oval is at A10. What are the names of three streets that are boundaries to the oval?

It's true!

After cars replaced horses as a means of transport, motoring became a popular leisure activity. People went for drives in the country on the weekends. They got their road maps at no charge from service stations where they stopped for fuel.

SYMBOLS

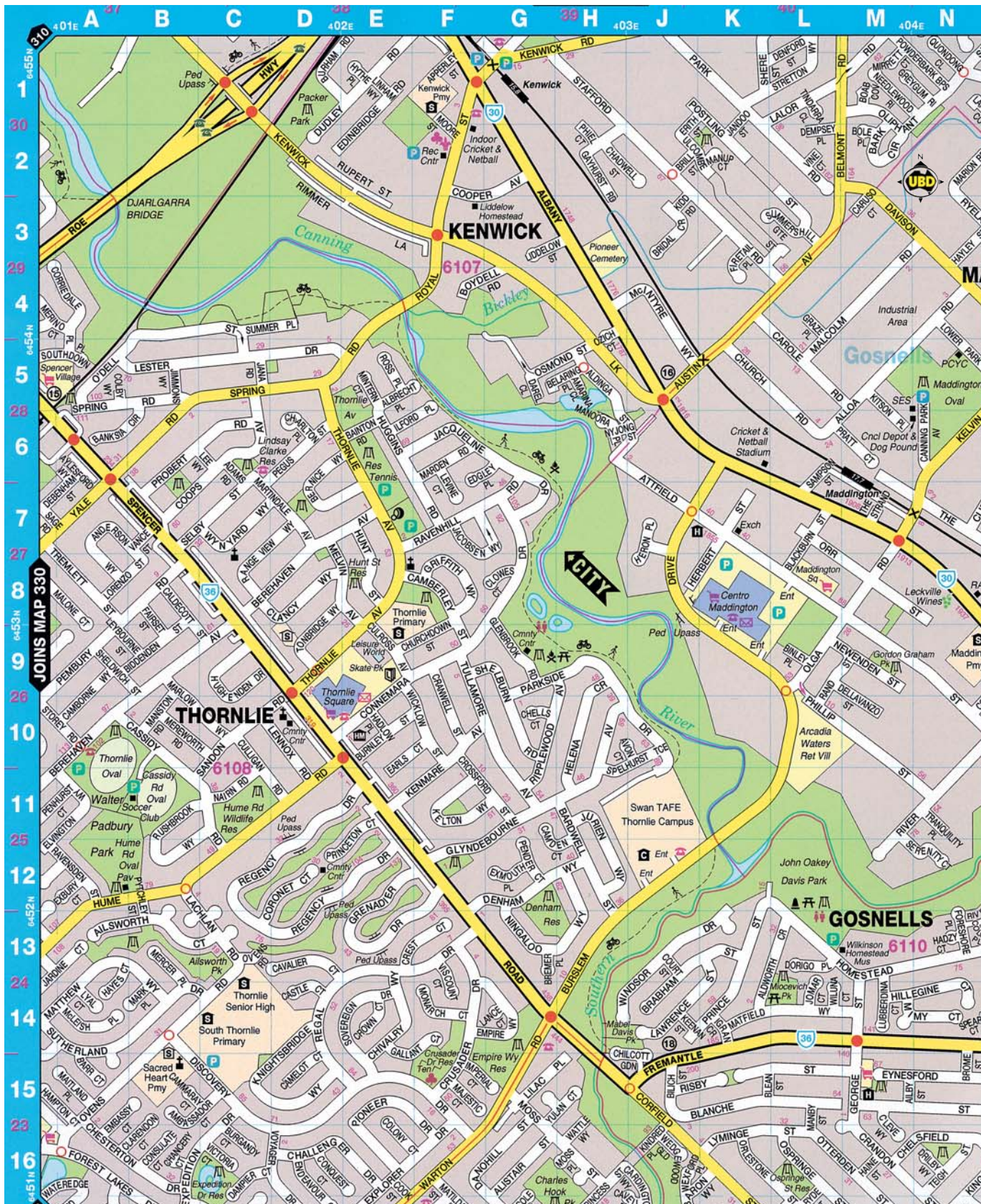
EXPLANATION OF MAP SYMBOLS



	Ambulance Station
	Barbecue
	Boat Fuelling Point
	Boat Ramp
	Bowling Club/Green
	Bush Fire Brigade
	Bus Stop
	Camping Area
	Caravan Park
	Car Park
	College – Private
	– Public
	Council Office
	Cycleway
	Emergency Phone
	Fire Station
	Golf Course
	Guides
	Hospital
	Hotel
	Information Centre
	Library
	Lookout – 360° view
	– 180° view
	Masonic Centre
	Monument/Memorial
	Motel
	Picnic Area
	Place of Interest
	Place of Worship
	Playground
	Police Station
	Post Office
	Quarry
	School – Private
	– Public
	Scouts
	Shopping Centre
	State Emerg. Service
	Swimming Pool
	Telephone
	Toilets
	Weighbridge
	Wineries

Freeway or Motorway
Proposed Freeway
Highway or Main Traffic Route & Footbridge
Alternative Traffic Route
Trafficable Road
Un-trafficable Road/Proposed Road
Traffic Light, Roundabout & Red Light Camera
No Left Turn, No Right Turn & Level Crossing
Road and Railway Bridges
One-way Traffic Route
National, State & Tourist Route Numbers
Direction of Perth City Centre
Distance by Road From GPO
House or Building Number
Railway Line with Station (distance from Perth)
Suburb Name
Postcode Number
Suburb Boundary
Unofficial Suburb Boundary
Locality Name
Local Government – Name
– Boundary
Ferry Route
Walking Tracks, Equestrian Trail and Cycleways
Park, Reserve, Golf Course, etc.
School or Hospital
Caravan Park, Cemetery, Shopping Centre, etc.
Buildings
Mall, Plaza
Wetlands
Land Subject to Inundation





4.11

Directions and distances

UBD street maps show directions on the map with the symbol .

In the Perth UBD street directory, north is always at the top of the page. The maps can have different scales. For the map on the facing page, the scale is: 1 cm represents 200 m.

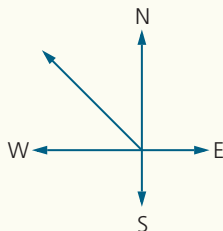
Worked example 4.11

Trent drove along Spencer Rd from the corner of Burslem Drive (G14) to Thornlie Ave (D10).

- In what direction did he drive?
- How far did Trent drive from G14 to D10?

Think

- The UBD symbol shows the compass directions. Trent drove in the direction that is in the middle of west and north.



This direction is called north-west.

- I can measure the distance on the map with a ruler, or a piece of string if the road isn't straight. Then I can use the scale that 1 cm represents 200 m.

Write

Trent drove north-west along Spencer Rd.

The distance on the map is 7.5 cm.

$$\begin{aligned}\text{True distance} &= 7.5 \times 200 \text{ m} \\ &= 1500 \text{ m} \\ &= 1.5 \text{ km}\end{aligned}$$

Exercise 4.11

Use the map on the facing page to complete this exercise.

- In what direction, **as the crow flies**, is it from the cricket and netball stadium at K6 to the following locations?
 - Dog pound (N6)
 - Centro Maddington (K8)
 - Tennis courts (E7)
 - Hume Rd Wildlife Reserve (C11)
 - Indoor cricket and netball (F2)
 - PCYC (N5)
- How far is it, as the crow flies, from Thornlie Square to Maddington station? Remember, 1 cm on this map represents 200 m.
- Calculate the distance along Albany Hwy from the traffic lights at Austin Ave (J5) to the lights at Royal St (F1).

HINT!

The expression 'as the crow flies' means in a straight line without worrying about travelling along roads.



4 Trent parked in Connemara Drive when he picked up a parcel at the post office in Thornlie Square (E9). He turned right into Spencer Rd, then right into the second street on the right. He turned right again into the first street on the right. His delivery was to a building on the right that is marked on the map.

- a** At what type of building is Trent delivering a parcel?
- b** How far did Trent drive from the post office to deliver the parcel?
- c** It took Trent $\frac{1}{4}$ hour to deliver the parcel. Calculate his average speed.

d Do you think Trent drove directly from the post office to make the delivery? Give a reason for your answer.

5 Trent has to pick up a parcel from Swan TAFE Thornlie Campus (J11) to deliver to a private address in Bainton Rd (E6).

- a** In what compass direction is Bainton Rd from the TAFE?
 - b** Write directions that Trent can follow to drive from the TAFE to Bainton Rd.
 - c** Calculate the distance Trent will drive from the TAFE to Bainton Rd.
- 6** When Trent was delivering a parcel in Bainton Rd, his boss phoned him and told him that he needed to go to Kenwick station (G1) to pick up some urgent parts that are required in the industrial area (M4).
- a** Describe the route that Trent could follow to go from Bainton Rd to Kenwick station.
 - b** What is the most direct way to travel from Kenwick station to Davison St opposite Hayley St (N3)?
 - c** How far is it from Bainton Rd to Davison St via Kenwick station, using your route in parts **a** and **b**?

It's true!

In large, traffic-congested cities, companies use bicycle couriers because bicycles don't get held up in the traffic. Often, bicycle couriers can deliver parcels more quickly than delivery trucks.

Practical activity: Online direction finding

To complete this activity you will need access to the Internet.

What you have to do

- 1** Log onto the site www.whereis.com.au.
- 2** Obtain directions from Bainton Rd to Kenwick station, then from Kenwick station to Davison St.
- 3** Compare the directions to the routes you described in question **6**.
- 4** According to the website, how long does it take to travel from Kenwick station to Davison St?

Quiz

How did you go?

- 4.02** 1 Justin is an apprentice panelbeater. He earns \$9.72 per hour. How much does he earn for working a 31-hour week?
- 4.02** 2 Alison earned \$345.80 for working 28 hours. How much does Alison earn per hour?
- 4.02** 3 Ngarie's annual salary is \$59 800. Calculate her fortnightly pay.
- 4.03** 4 Brad's normal pay is \$11.20 per hour. When he works overtime he is paid at time-and-a-half. How much will he be paid for working 6 hours overtime?
- 4.03** 5 Belinda works a 36-hour week and she earns \$13.60 per hour normal time. If she is required to work overtime, she is paid time-and-a-half for the first 4 hours overtime and double-time after that. Last week Belinda worked a 43-hour week. Calculate her pay for last week.
- 4.03** 6 Nicole works in the pro shop at the golf club. Her basic wage is \$23.10 per hour. She is employed to work a 7-hour day, Monday to Friday. Any overtime she works on a weekday is paid at time-and-a-half, and she is paid double-time for all work on weekends.

Last week Nicole worked:

Monday	7 hours
Tuesday	9 hours
Wednesday	5 hours
Thursday	11 hours
Friday	7 hours
Saturday	5 hours

- a** How many hours does Nicole normally work each week?
- b** How many hours overtime did Nicole work last week?
- c** How much does Nicole normally earn each week?
- d** How much did Nicole earn last week?
- 4.03** 7 Trent drives a delivery truck. He is paid \$8.90 per hour plus 40c for every kilometre he drives. Calculate Trent's pay for a week when he worked for 37 hours and drove 378 km.
- 4.04** 8 Guy has a part-time job as a labourer. He earns \$14.10 per hour. If he is required to work in wet, muddy conditions, he receives an additional 85c per hour. How much will Guy earn for working 7 hours, of which 3 hours were in wet, muddy conditions?



4.05

9 Melanie sells make-up and perfume on commission. She is paid a weekly retainer of \$260 and she receives 5% commission on all items she sells. Calculate Melanie's pay for a week when her sales totalled \$950.

4.05

10 Judith sells real estate. She receives 2% of the selling price of all properties she sells. Calculate the commission Judith will earn for selling a house for \$1.4 million.



4.06

11 Employers are required to pay an amount equivalent to 9% of an employee's wage or salary into superannuation for the employee. Joslyn earns \$580 per week. What is the minimum amount her employer is required to pay into superannuation on her behalf?

4.07

12 Bruce's weekly pay is \$720. Each week his employer deducts \$85 for PAYG tax and \$55 for loan repayments. Calculate Bruce's net or take-home pay.

4.07

13 Stephan pays \$127 in tax each week. How much tax will he pay in a year?

4.08

14 Jamie paid \$5690 in PAYG tax. At the end of the year, he calculated that he should have paid \$5970 in income tax. Will Jamie get a tax refund? Explain your answer.

4.08

15 Faiz's taxable income is \$73 600.

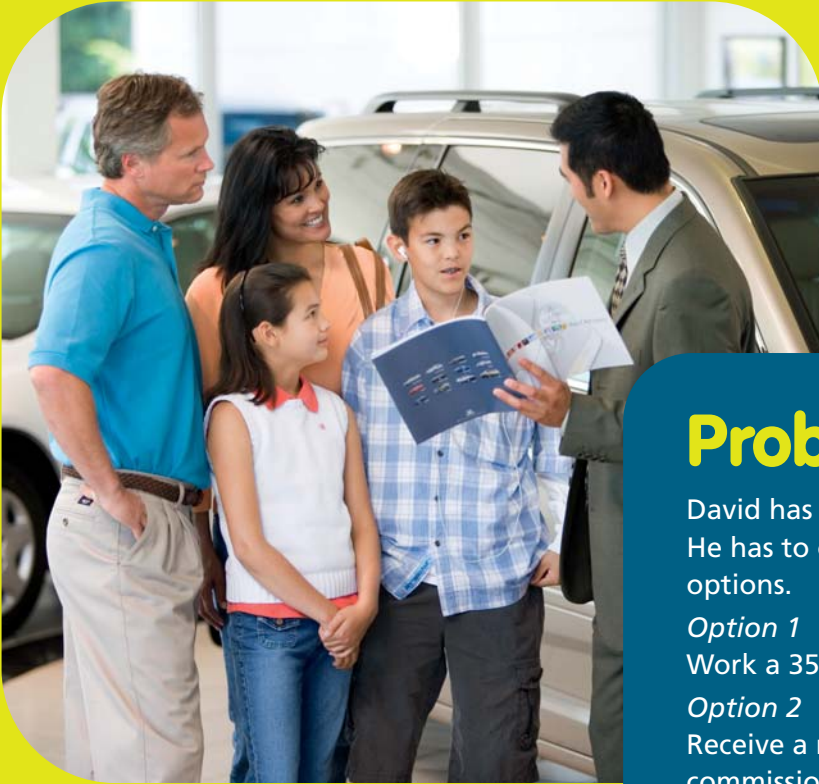
- a** Use the tax table on page 116 to determine the amount of income tax Faiz is required to pay.
- b** The Medicare Levy is 1.5% of taxable income. Calculate Faiz's Medicare Levy.
- c** How much of Faiz's taxable income will remain after he has paid his income tax and Medicare Levy?

4.10

16 Use the map on page 124 to answer the following questions.

- a** What sport is played in the park located at E7?
- b** What are the position coordinates of the south-west on-ramp to the Roe Hwy at the top left of the map?
- c** The scale on the map is 1 cm represents 200 m. How far is it along the walking track from the northern corner of the TAFE at J11 to where the track meets Royal Rd?
- d** Write directions for travelling from Thornlie Senior High School (C14) to the Leisure World Skate Park (E9).

4.11



Problem

David has just been offered a job selling cars. He has to choose between two different pay options.

Option 1

Work a 35-hour week at \$12.60 per hour.

Option 2

Receive a retainer of \$240 per week plus 2% commission on the total value of the weekly sales he makes.

Which pay option should David accept?

Solution

Option 1

$$\begin{aligned}\text{Weekly pay} &= 35 \times \$12.60 \\ &= \$441\end{aligned}$$

Option 2

$$\begin{aligned}\text{Pay if he sells cars worth \$10 000} &= \$240 + \frac{2}{100} \times \$10\,000 \\ &= \$440\end{aligned}$$

$$\begin{aligned}\text{Pay if he sells cars worth \$20 000} &= \$240 + \frac{2}{100} \times \$20\,000 \\ &= \$640\end{aligned}$$

Each week, David needs to sell cars worth more than \$10 000 for option 2 to be better than option 1. David should make his choice depending on the value of the cars he thinks he can sell per week.